

In due course, the position-dependent-mass jump of (1) implies

$$q(x) = x \sqrt{f(h(x))} = x [1 + \mu h(x)], \quad (17)$$

and

$$U(x) = \frac{\mu}{1 + \mu h(x)} \implies U(0) = \frac{\mu}{1 + \mu/2}. \quad (18)$$

It should be noted here that, in a straightforward manner, one may substitute $U(0)$ in (18) to obtain the transmission and reflection coefficients (15) and (16), respectively, as

$$R = -\frac{4\mu(1 + \mu/2)^2}{4(1 + \mu/2)^4 + \mu^2}, \quad (19)$$

and

$$T = \frac{4(1 + \mu/2)^4 - \mu^2}{4(1 + \mu/2)^4 + \mu^2}. \quad (20)$$

Consequently and asymptotically speaking, a Taylor series expansions about $\mu \rightarrow 0$ (i.e., $0 < \mu \ll 1$, also very likely experimentally feasibly applicable) would result in casting the reflection and transmission intensities, respectively, as $|R|^2 \approx \mu^2 + O(\mu^3)$ and $|T|^2 \approx 1 - \mu^2 + O(\mu^3)$. Within such asymptotic tendencies, it is obvious that increasing the value of μ from just above 0 to 1 would make the derivative-of-the-Dirac's delta scatterer less transparent until total reflection takes place when $\mu = 1$.

3 Concluding remarks

In this work, a quasi-free quantum particle (i.e., $V(x) = 0$ whereas $V_{eff}(q(x)) \neq 0$) endowed with a position-dependent-mass jump (1) is considered. Whilst, under Mustafa's and Mazharimousavi's ordering ambiguity parametrization (i.e., $\alpha = \gamma = -1/4$ and $\beta = -1/2$) the free PDM quantum particle remained free, we have witnessed (only under a specific though rather new set of ordering am-